

10(a)	any nine from: 1 Na^+ / sodium ion, channels open ; 2 Na^+ / sodium ions, enter, neurone / axon / (nerve) cell ; 3 membrane (potential) becomes, positive / +40 mV / depolarised ; 4 (repolarisation / after peak) Na^+ / sodium ion, channels close ; 5 K^+ / potassium ion, channels open ; 6 K^+ / potassium ions, leave / move out (of cell) ; 7 membrane (potential) becomes, negative / –90 mV / repolarised ; 8 local circuits ; 9 myelin sheath / Schwann cells, insulates / stops ion movement ; 10 action potential / depolarisation, only occurs at nodes of Ranvier ; 11 saltatory conduction ; 12 fast transmission (of, action potential / impulse) ; 13 one-way transmission (of, action potential / impulse) ; 14 AVP ;	9
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Question	Answer	Marks
10(b)	<i>any six from:</i> 1 maintain / keep / regulate, internal, environment / conditions ; 2 within, narrow / set, limits or around, optimum value / set point / norm ; 3 low (body) temperature and consequence ; 4 high (body) temperature and consequence ; 5 low blood glucose (concentration) and consequence ; 6 high blood glucose concentration and consequence ; 7 low, (blood) water potential / ψ, and consequence ; 8 high, (blood) water potential / ψ, and consequence ;	6